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FIN 302 Finance and Markets

Why Financial Crises?

By Dr. Shan Wu

Chapter Preview (1 of 3)

- Financial crises are major disruptions in financial markets characterized by sharp declines in asset prices and firm failures.
- Beginning in August 2007, the U.S. entered into a crisis that was described as a “once-in-a-century credit tsunami.”
- An application of the economic analysis of the effects of **asymmetric information** (**adverse selection** and **moral hazard**).



Chapter Preview (2 of 3)

In this chapter, we'll learn...

- Why did this financial crisis occur?
- Why have financial crises been so prevalent?
- What insights do they provide on the current crisis?
- Why are financial crises almost always followed by severe contractions in economic activity?

Chapter Preview (3 of 3)

In this chapter, we develop a framework to understand the dynamics of financial crises. Topics include:

- What Is a Financial Crises
- Dynamics of Financial Crises in Advanced Economies

What Is a Financial Crises?

- We've discussed how a functioning financial system is critical to a robust economy.
- However, both **moral hazard** and **adverse selection** are still present. The study of these problems is the basis for understanding and defining a financial crisis.
- If capital goes to the wrong uses or does not flow at all
 - The economy will operate inefficiently or go into an economic downturn.

To understand what is a financial crises...

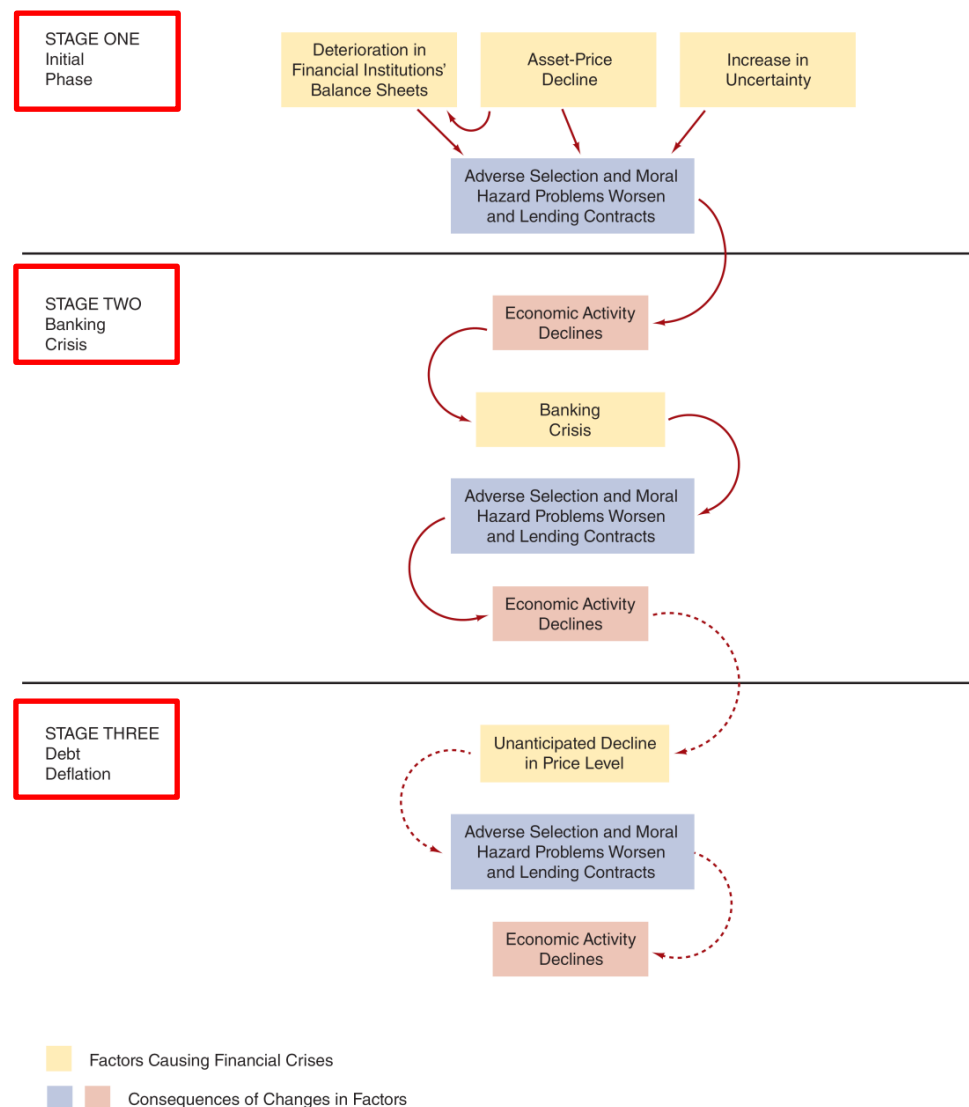
- Asymmetric information
 - How asymmetric information problems generate adverse selection and moral hazard problems.
- Asymmetric information creates barriers between savers and firms with productive investment opportunities.
- Signal for high asymmetric information
 - High financial frictions.
 - High credit spread
 - the difference in yield between a treasury (a risk-free) and corporate bond of the same maturity.

To understand what is a financial crises...

- Signal for high asymmetric information
 - High financial frictions.
 - High credit spread
 - $\text{Credit spread} = \text{Yield on a corporate bond} - \text{Yield on a government bond}$
 - the extra yield that investors demand for holding a risky bond instead of a safe bond
- When financial crises come, how do financial frictions and credit spreads change?
 - In a crisis, uncertainty rises, lenders become cautious, and risky firms must pay much more to borrow.
 - In normal times, firms can borrow relatively easily.

Dynamics of Financial Crises in Advanced Economies

- Financial crises happen so often, and each event helps economists gain insights into present-day turmoil.
- These crises usually proceed in 2 or 3 stages.

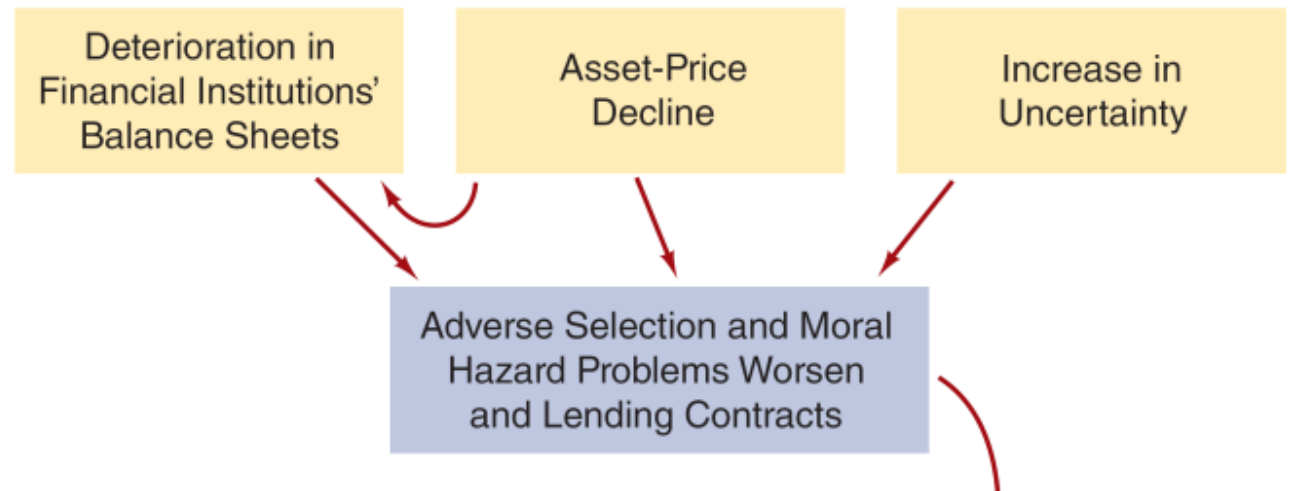


Stage One: Initial Phase (1 of 6)

Financial crisis can begin in several ways:

1. Credit Boom and Bust
2. Asset-Price Boom and bust
3. Increase in Uncertainty

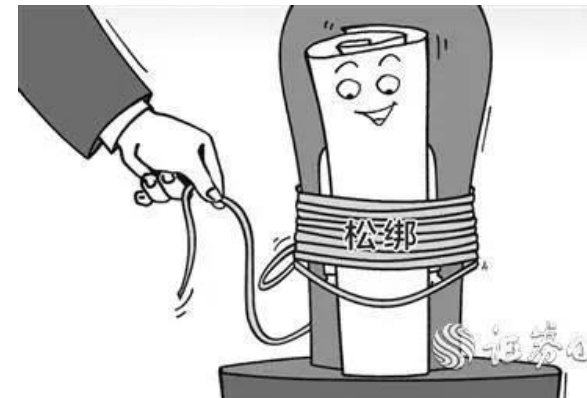
STAGE ONE
Initial
Phase



Stage One: Initial Phase (2 of 6)

1. Credit Boom and Bust

- Happens when banks give out a lot of loans easily (the **boom**), and then suddenly stop lending when they realize too much borrowing has led to financial problems (the **bust**).
- **Boom Phase:**
 - Banks give out loans easily, even to risky borrowers.
 - People and businesses borrow a lot to buy houses, stocks, or expand businesses.
 - Prices of assets (like houses and stocks) go up fast.
 - Everyone feels rich and confident.



Stage One: Initial Phase (2 of 6)

1. Why Credit Boom?

- The seeds of a financial crisis can begin with mismanagement of **financial liberalization or financial innovation**:
 - elimination of restrictions.
 - Introduction of new types of loans or other financial products (e.g. derivatives)
- Banks make risky, high-interest loans to borrower-spenders.
- **Huge profits** makes banks more risk-taking



Stage One: Initial Phase (2 of 6)

1. The seeds of a financial crisis can begin with mismanagement of **financial liberalization or financial innovation**:

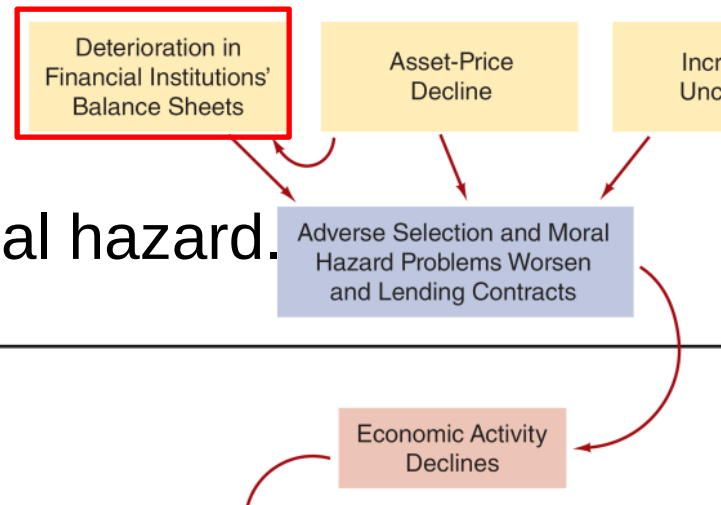
- Consequence ? Why business close?
 - Lenders may not have the expertise, or the incentives to manage risk of these innovative products, as long as they are highly profitable
- Risk management is lacking → Credit busts
- Credit busts
 - Borrowers struggle to pay back loans.
 - Banks panic and stop lending.
 - Asset prices crash because people can't afford to buy.



Stage One: Initial Phase (3 of 6)

1. Credit **Bust** from banks perspective: Loans availability ↑ ↓
- Loan losses mount, and value of loans fall, leading to a reduction in capital.
 - Financial institutions cut back in lending, a process called **deleveraging**
 - Banking balance sheet deteriorated → Depositors ...
 - → Loans amount...
 - Loans become scarce →
 - Less incentive to evaluate firms.
 - Increased adverse selection and moral hazard.

STAGE ONE
Initial
Phase



STAGE TWO
Banking
Crisis

Summary:

First trigger-Cred

Financial crisis can begin in several ways:

1. Credit Boom and Bust

- Happens when banks give out a lot of loans easily (the **boom**), and then suddenly stop lending when they realize too much borrowing has led to financial problems (the **bust**).

<https://mp.weixin.qq.com/s/DmDcAp42CMKnibOK51c9OA>

JFE: 为何信贷繁荣会导致严重的金融危机以及随后缓慢的经济复苏

Original 唧唧堂 唧唧堂 2024年6月3日 21:53 上海

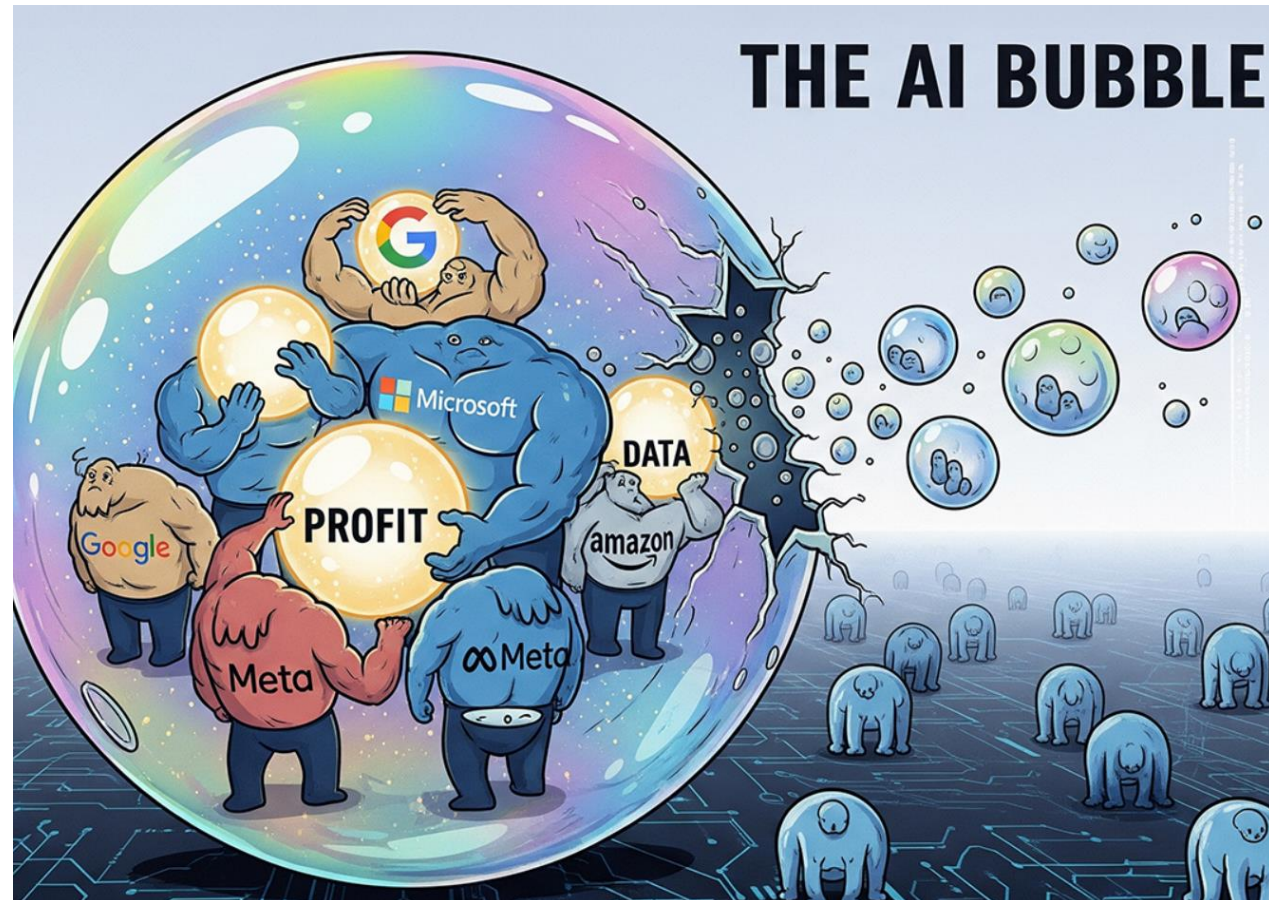


本期推荐一篇2024年2月发表在JFE上的论文《延迟危机和缓慢复苏》。2007-2009年的全球金融危机引发了对信贷周期和金融危机循环的广泛讨论和研究。实证研究表明，信贷繁荣往往预示着金融危机的到来，且信贷繁荣的规模可以预测随后的危机严重程度以及经济衰退的持续时间。然而，从理论角度来看，仍然存在一些自然问题，例如：如果经济主体（企业、家庭或金融中介）意识到前方的巨大风险，为什么他们还会承担如此多的债务（信贷）？为什么信贷繁荣通常会在低效情况下持续很长时间然后崩溃？

Stage One: Initial Phase

Financial crisis can begin in several ways:

1. Credit Boom and Bust
2. Asset-Price Boom and bust



Stage One: Initial Phase (5 of 6)

2. Asset-Price Boom and bust

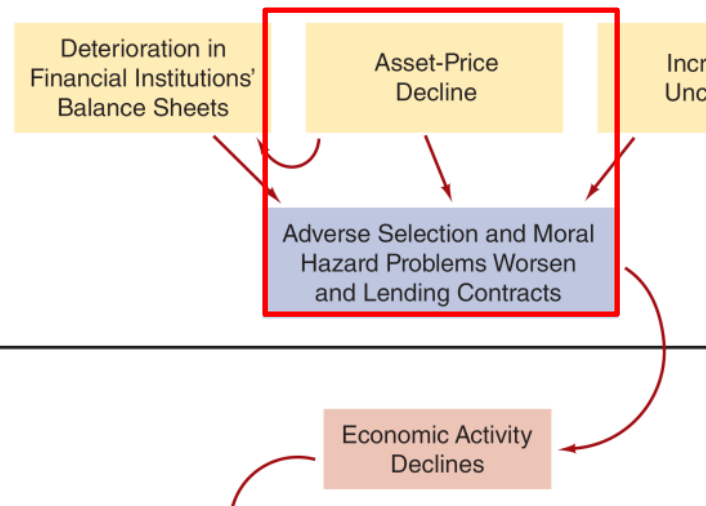
- Asset-price bubble: asset values $>$ their fundamental values. When does it tend to have?
- Often driven by credit booms.
- When the bubble bursts, what happens to firms' net worth?
- How does it change firms' risk preference?
- How about moral hazard?
- How do FIs react?

Price bubble bursts →

Adverse selection and Moral hazard

→ Less lending

STAGE ONE
Initial
Phase



STAGE TWO
Banking
Crisis

Stage One: Initial Phase (5 of 6)

2. Asset-Price Boom and bust

- How does pricing bubble influence banks?
 - What happen to the value of FIs if **bad loans** increase?
 - Also see a fall in **FIs' assets**
 - leading again to deleveraging.

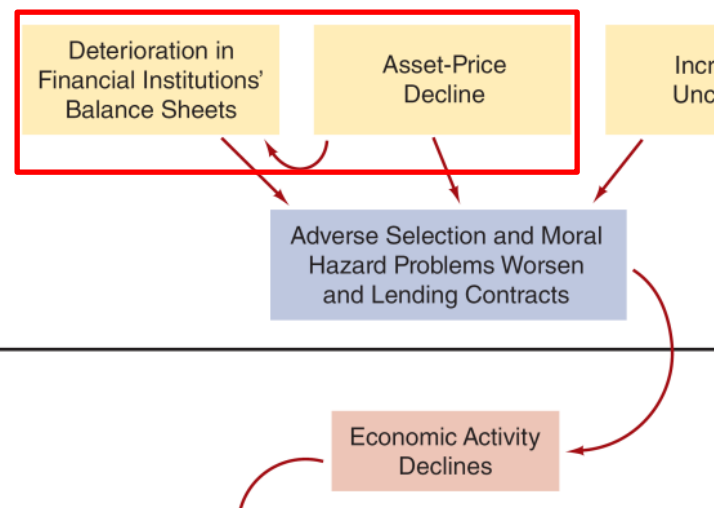
Price bubble bursts →

Value of firms' assets decline

Value of FIs' assets decline →

FIs' balance sheets deteriorated

STAGE ONE
Initial
Phase



STAGE TWO
Banking
Crisis

Asset-Price Boom and bust

- How to accurately measure if there is a bubble?
- Asset-price bubble: asset values $>$ **fundamental values**
- Why bubbles?
- Bubbles result from individual-level misperceptions. (Behavioral finance)
- Bubbles are social phenomena and reflect contagion of emotions as well as ideas. (Social finance)
 - Bubbles derive from the transmission of financial ideas, feelings, and behaviors through conversation, mass media and social media, and personal observation.

Asset-Price Boom and bust

- Case: Chinese Housing Bubble
- Why there was a housing bubble in the beginning?
- Why Home prices are falling, developers are defaulting?

https://www.youtube.com/watch?v=dnP_MxXY9qs&t=612s

Stage One: Initial Phase

Financial crisis can begin in several ways:

1. Credit Boom and Bust
2. Asset-Price Boom and bust
3. Increase in Uncertainty



Stage One: Initial Phase (6 of 6)

3. Increase in Uncertainty

- Periods of high uncertainty can lead to crises, such as stock market crashes or the failure of a major financial institution.



Stage One: Initial Phase (6 of 6)

- 3. Increase in Uncertainty
- Periods of high uncertainty can lead to crises, such as stock market turbulence

Oil prices have risen sharply since the start of the Iran war

Brent crude, \$ a barrel (hourly)



Guardian graphic. Source: LSEG. Note: hourly data to 0800 GMT, 19 March 2026

Stage One: Initial Phase (6 of 6)

3. Increase in Uncertainty

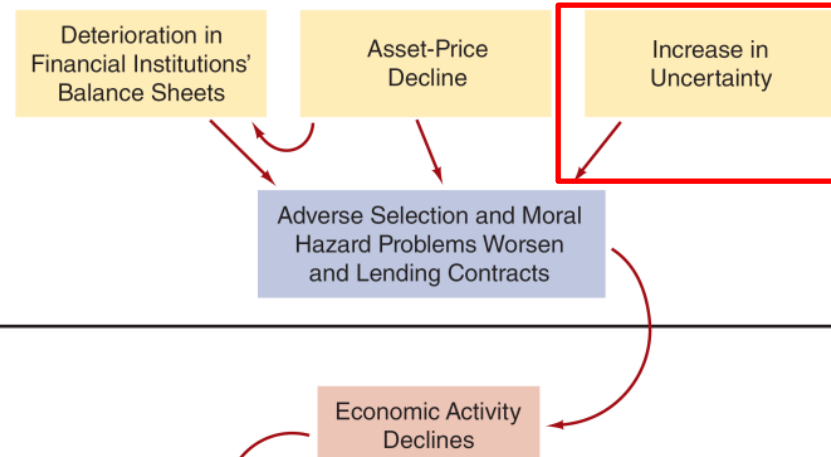
- Periods of high uncertainty can lead to crises, such as stock market crashes or the failure of a major financial institution.
- With information hard to come by, moral hazard and adverse selection problems....
- increase, reducing lending and economic activity

Uncertainty increases

→ Adverse selection, Moral hazard

→ Less lending

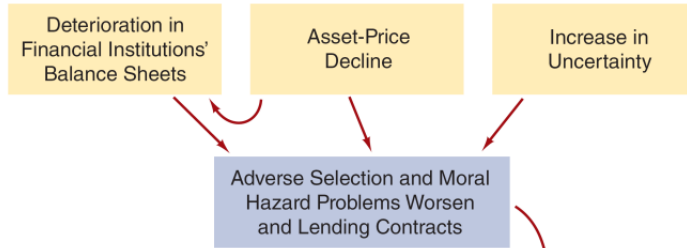
STAGE ONE
Initial
Phase



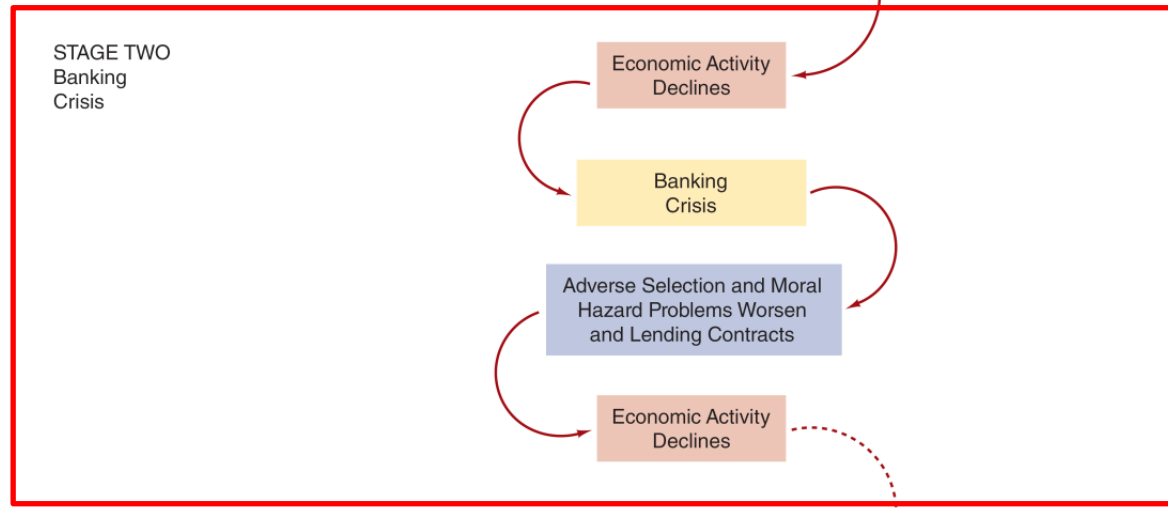
STAGE TWO
Banking
Crisis

Stage Two of Financial Crises : Banking Crisis

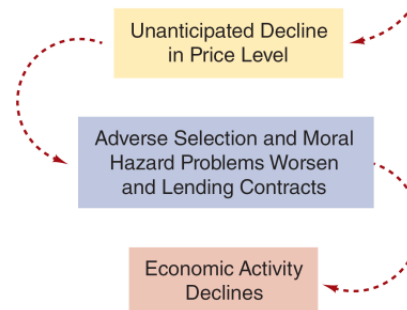
STAGE ONE
Initial
Phase



STAGE TWO
Banking
Crisis



STAGE THREE
Debt
Deflation



Factors Causing Financial Crises

Consequences of Changes in Factors

Stage Two: Banking Crisis

- If FIs' balance sheets keep deteriorating, what could happen?
- Financial institutions insolvency.
- A **bank panic** if too severe.
- Many depositors withdraw cash almost simultaneously. Why?
- A huge decline in depositor confidence or fear that the bank will be closed.



Case: Silicon Valley Bank

硅谷银行倒闭前被挤提420亿 机构客户为主九成存款 “未被保险”

Silicon Valley Bank suffers \$42 billion bank run before collapse, with most depositors being institutional clients and 90% of deposits "uninsured"

文 | 财新 王石玉

2023年03月11日 15:41



硅谷银行以服务创业企业著称，例如2022年中，科技与生物健康IPO公司中有四成是其客户，这意味着危机传导和管控将极其特殊，美国财长耶伦表示对银行业监管机构有能力采取适当行动“完全有信心 (full confidence) ”



- Bank panics were a feature of all U.S. financial crises, occurring every 20 years or so until World War II.
- ~60-70 countries have experienced significant banking crises since the early 1990s.

<https://www.youtube.com/watch?v=87YaO-F-xxA>

What is a bank panic?

- A widespread number of bank depositors suddenly try to withdraw their funds.
- Features
 1. **Loss of Confidence:** A sudden, widespread loss of confidence in banks → A rush to withdraw deposits.
 2. **Systemic Event:** not just a run on an isolated bank; affecting the entire banking system.
 3. **Suspension of Convertibility:** The demand for cash can be so overwhelming that banks may temporarily stop allowing depositors to convert their deposits into cash.

Stage Two: Banking Crisis

- Bank panic → Cash balances ...?
- Fall → Sell assets quickly (fire sales) → further deteriorating their balance sheet →
- Bank Insolvency →
- Fewer banks operating → Asymmetric information...?
- Adverse selection and moral hazard become severe →
- Finally, recovery
 - Insolvent firms shut down, uncertainty declines, stock market recovers and balance sheets improve.
 - It takes years for a full recovery

A signal for recession?

- The historical evidence suggests that panics often occur at or near the peak of a business cycle, just before a recession starts.
- So, panic may be a signal to broader economic fluctuations.

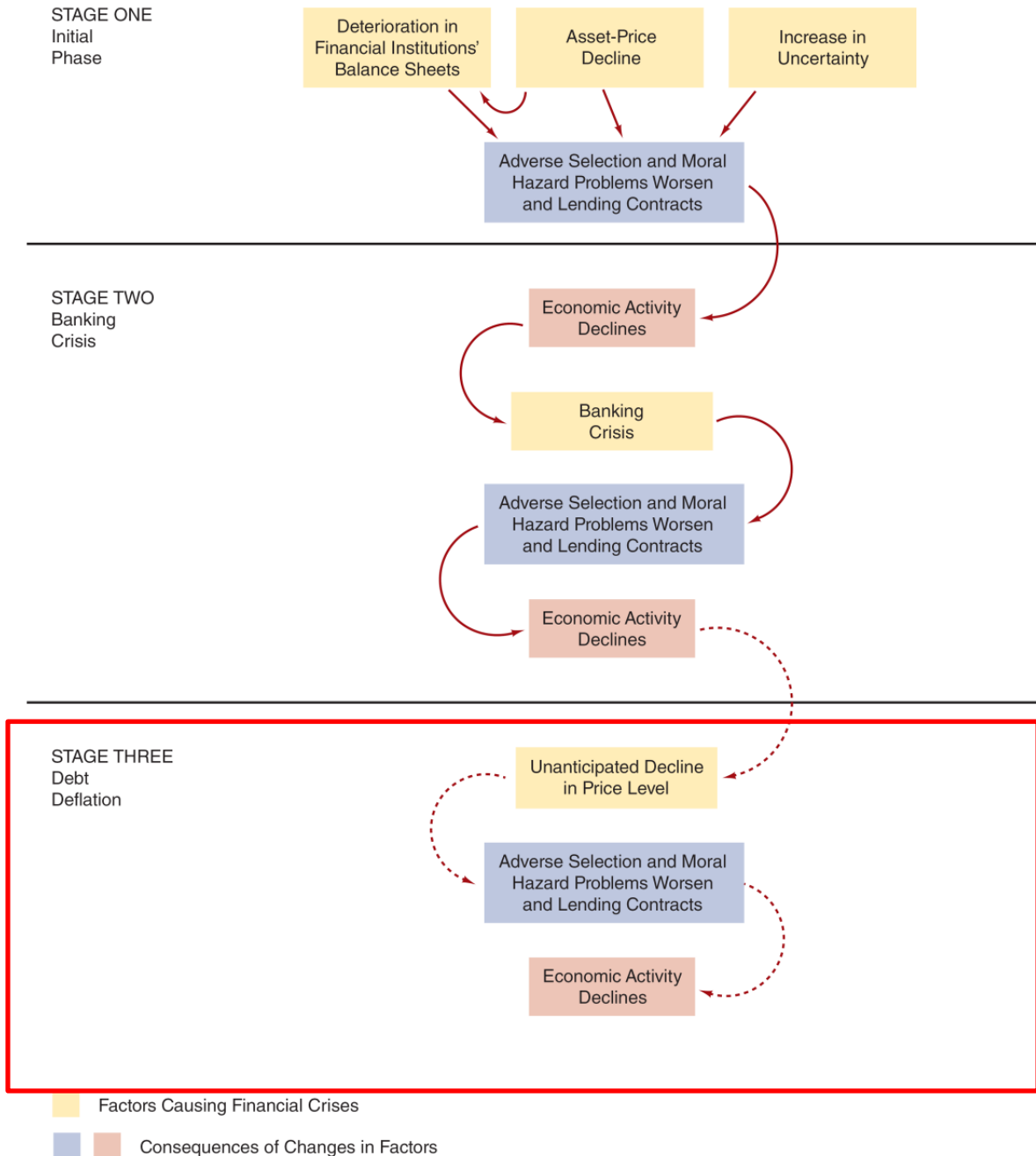
What kind of banks are more vulnerable to panic?

- Large banks or Small banks?
- The U.S. historically had numerous small, undiversified banks, which made it more susceptible to panics.
- In contrast, Canada had a system with a few large, well-branched banks, leading to greater stability and fewer panics.
- Systems with many small, undiversified banks tend to be more prone to panics, while systems with a few large, well-branched banks tend to be more stable.
- i.e. structure of the banking industry, may determine its vulnerability to panics.

Structural characteristics of banks in a country

- **Structural characteristics** of the banking industry, may determine its vulnerability to panics.
1. **The number and size of banks in the system:** Are there many small banks, a few large banks, or a mix of both?
 2. **The degree of concentration:** Is the market dominated by a few large players, or is it more fragmented with many smaller banks competing?
 3. **The extent of branching:** Allowed to operate multiple branches across different regions or are they restricted to a single location?
 4. **The presence of barriers to entry:** How easy is it for new banks to enter the market and compete with existing players?
 5. **The degree of competition:** How intense is the rivalry among banks for deposits and loans?

Sometimes, Stage Three: Debt Deflation



Stage Three: Debt Deflation

- **Debt deflation**

- If the crisis also leads to a sharp decline in prices
- Prices of goods, houses, and stocks are falling, while debt burden remains unchanged
- Resulting in the increase of debt burdens from borrowers' perspective.
 - Because Money is more valuable, compared with goods (Everyone is short of money, so the purchasing power of money increases)
- Opposite to Inflation as we normally experience

Stage Three: Debt Deflation

- **Debt deflation**
 - Resulting in the increase of debt burdens of borrowers.
 - If borrowers cannot repay, they may take high risk, for a chance of high return.
- This leads to an increase in adverse selection and moral hazard, which is followed by decreased lending
- Economic activity remains depressed for a long time

Recovery

- After Stage 2 or Stage 3
- Finally, recovery
 - Insolvent firms shut down, uncertainty declines, stock market recovers and balance sheets improve.
 - It takes years for a full recovery

Consequence

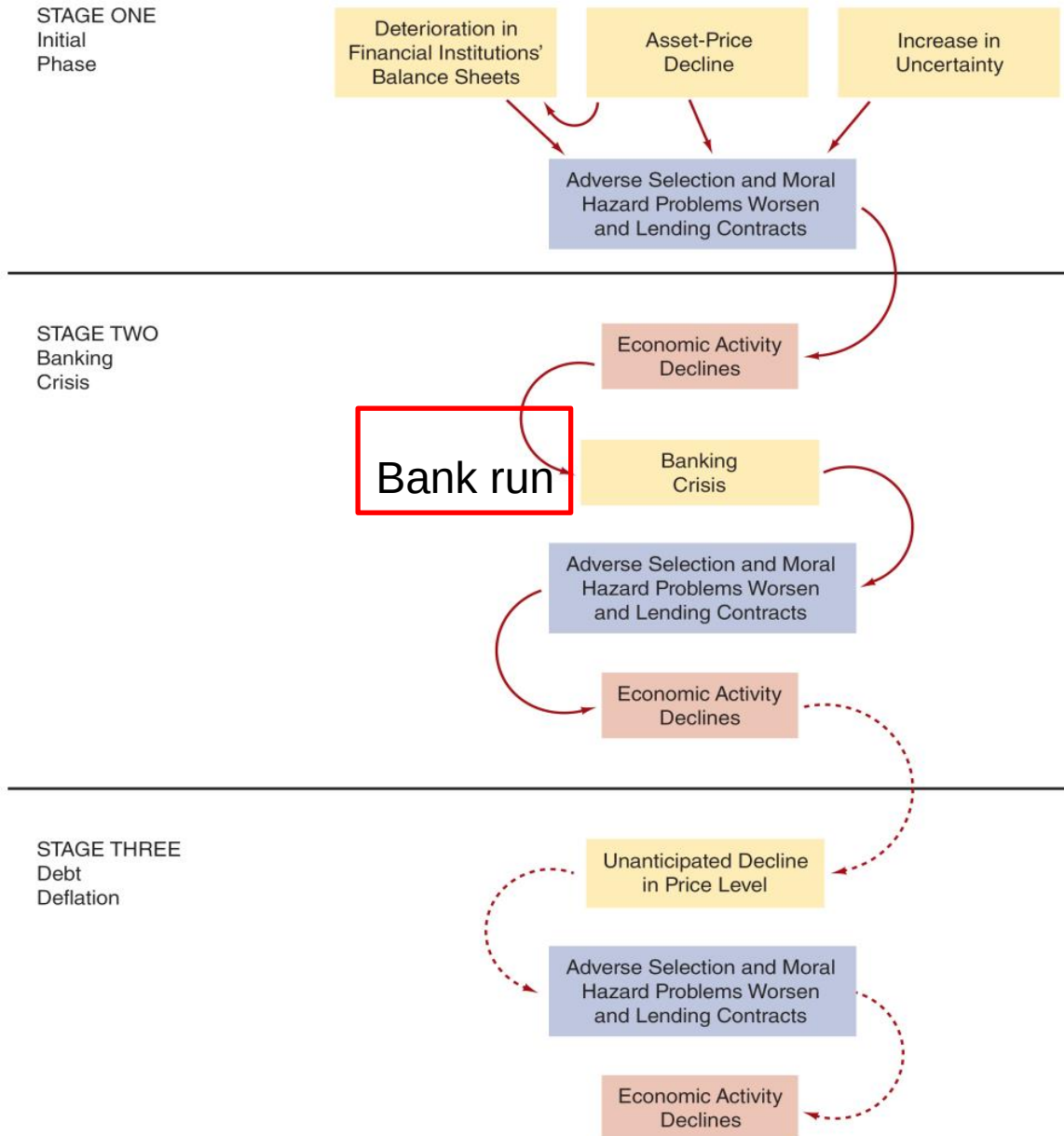
- Huge loss:
- Economic activity remain depressed for a long time
- Large job losses
- Social unrest



Summary

Dynamics of Financial Crises

- Credit Boom and Bust
- Asset-Price Boom and bust
- Increase in Uncertainty

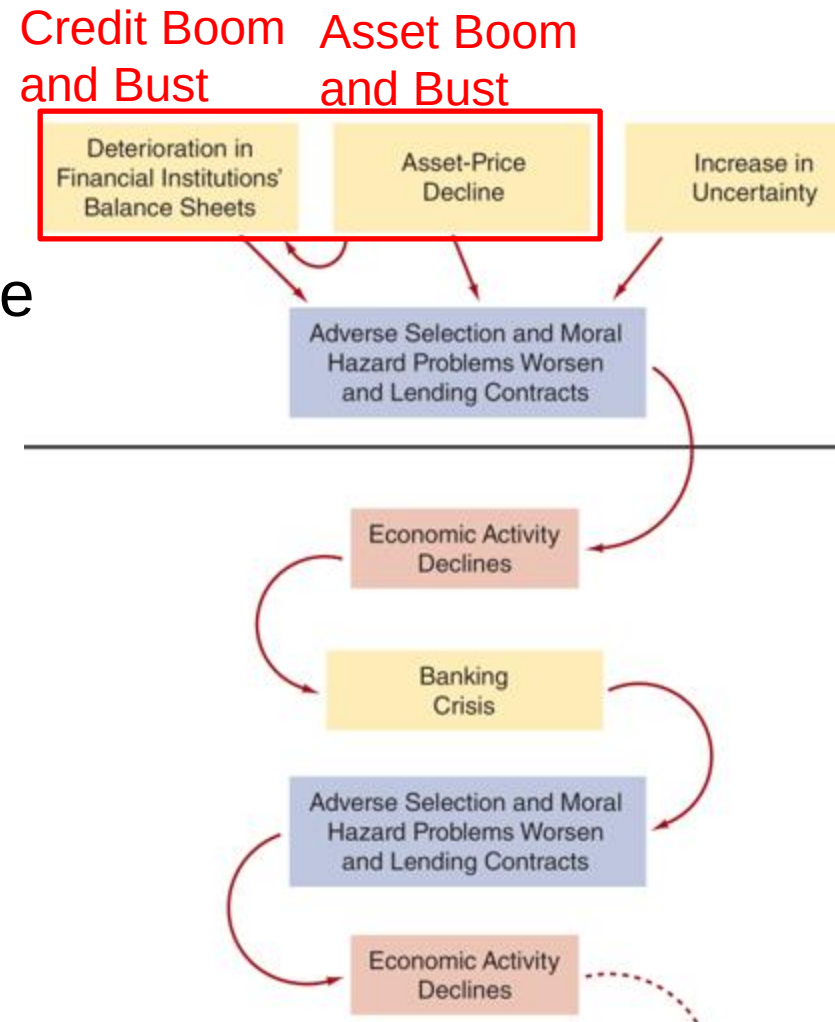


Case:

The Global Financial Crisis of 2007–2009

We begin our look at the 2007–2009 financial crisis by examining three central factors:

1. Financial innovation in mortgage markets
2. Agency problems in mortgage markets
3. The role of asymmetric information in the credit rating process



1. Financial innovation in mortgage markets

- Before 2000, only credit-worthy(**prime**) borrowers get residential mortgages
- Less-than-credit worthy borrowers can purchase homes through mortgage loans, a practice almost nonexistent until the 2000s. Why?
- After 2000, advances in computer technology
- New statistical techniques
 - Cheaply quantify the default risk of potential borrowers, as long as they have credit records.
 - Transaction costs are reduced, securitization emerged
 - less credit-worthy borrowers also get mortgages:
Subprime mortgages
 - Subprime Mortgage Crisis

Financial innovation in mortgage markets

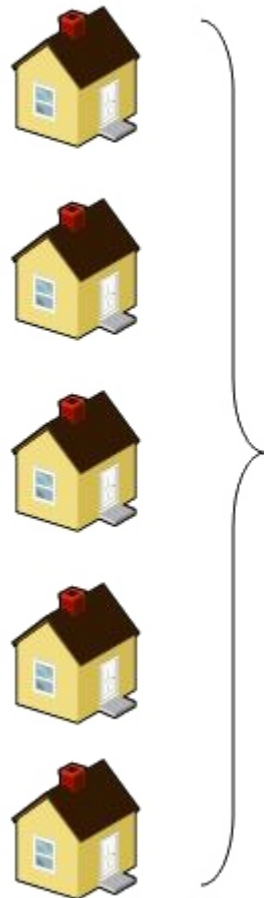
- Financial engineering further enhance and distribute risk
- Develop new, complex financial products.
- Designed to have different risks for investors with different risk preferences, i.e. Structured credit products.
- What's the potential risk?
 - Too sophisticated
 - Cannot actually estimate the real risk of these products, even for people who invent them



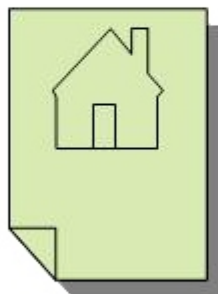
E.g. Financial innovation in mortgage markets

- Mortgage-backed security (MBS) ≈ Bond
 - Consists of a bundle of home loans debt
 - Investors receive periodic payments similar to bond

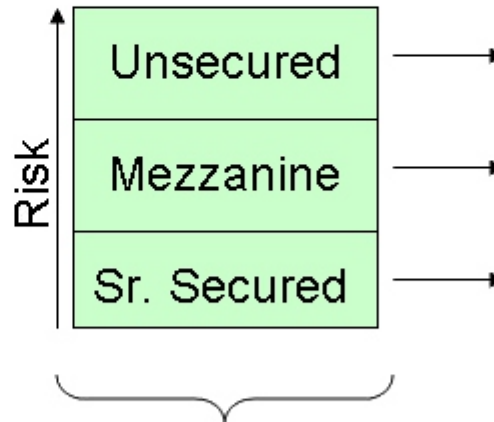
Mortgages



Mortgage
Pool



Mortgage Backed Security
a.k.a. “Collateralized Debt Obligation”
(divided into “tranches”
and sold at dif. levels of risk)



a.k.a. “Derivatives”
(broken up into groups and ‘derived’
from the original pool)

Expected
Return

12%

9%

7%

- Bank is an intermediary between homebuyer and investors.
- The bank handles the loans and then sells them at a discount to investors.
- For the investor, an MBS is safe (mortgage loans back them up).
- But the real risk?